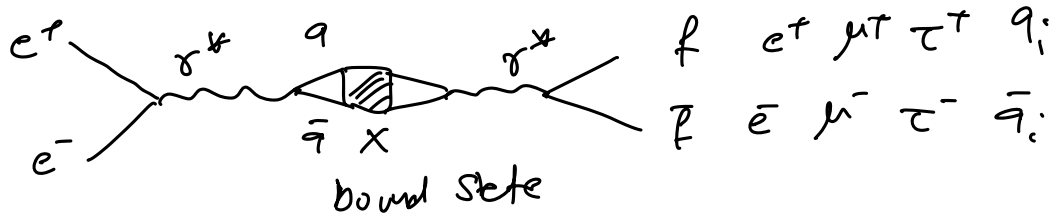


Direct evidence for quarks

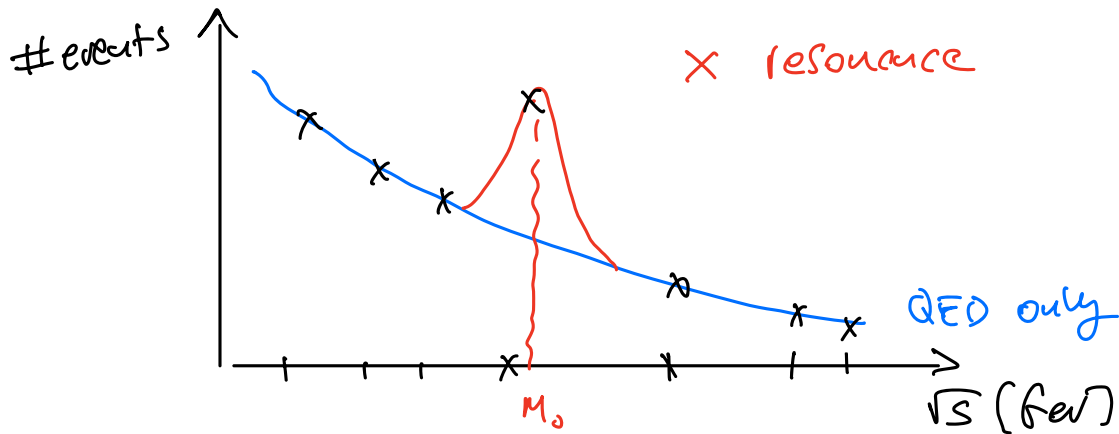
1) $e^+e^- \rightarrow X = q\bar{q} \rightarrow \mu^+\mu^-, e^+e^-$



a) look at cross section σ vs. $\sqrt{s} = E^+ + E^- = 2E_{beam}$

$\sigma(e^+e^- \rightarrow \mu^+\mu^-) \propto \frac{1}{s}$ QED w/ no resonance.

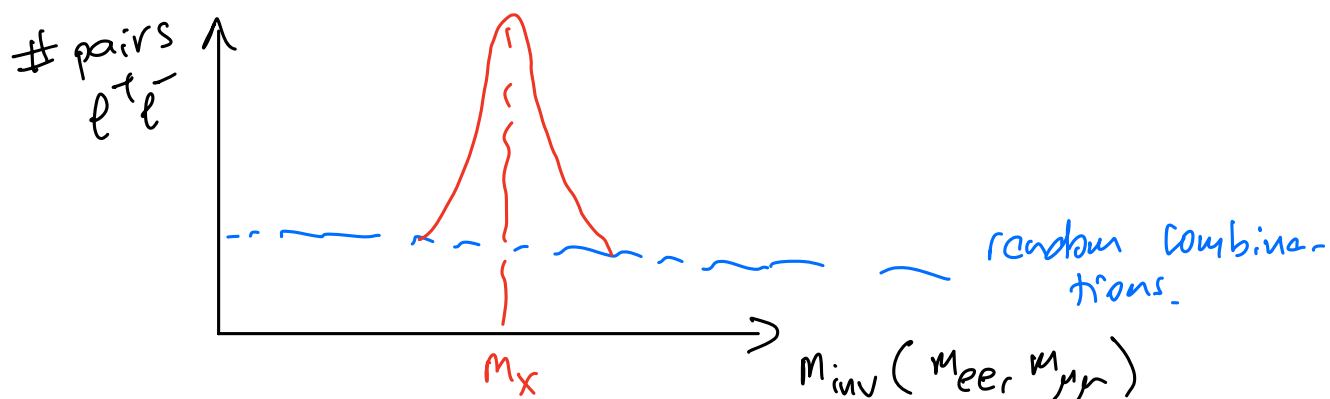
$\sigma(e^+e^- \rightarrow X \rightarrow \mu^+\mu^-) \propto \frac{1}{s} \frac{\Gamma^2}{(X - m_0)^2 + (\Gamma/2)^2}$ BW Structure



counting experiment $\cap$ different $\sqrt{s}$

b) look at invariant mass $M_{e^+\mu^+}$ ($M_{\mu^+\mu^-}$)

$\Rightarrow$ requires reconstruction of $\vec{p}_i E$ for tracks



leptonic final states expected signal/bkg ratio.

2) hadronic production $p + \text{target} \rightarrow \text{hadrons (X + pions + baryons)}$

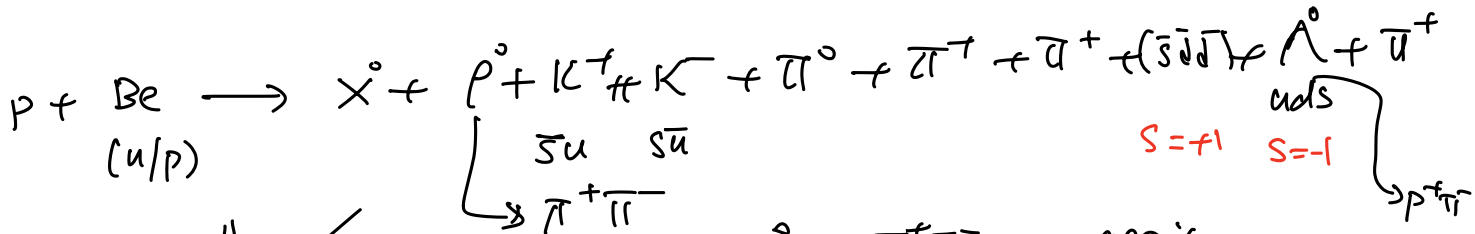
Be
Cu

$\rightarrow e^+e^-$

$\ell = e, \mu$

typically many hadrons
in final state

look at invariant mass $\ell^+\ell^-$ search for peaks



Baryon # ✓

elect. charge ✓

$$\rho^0 \rightarrow \pi^+\pi^- \sim 100\%$$

$$\pi^\pm \rightarrow \ell^\pm (\nu_\ell / \bar{\nu}_\ell)$$

$$K^\pm \rightarrow \ell^\pm (\nu_\ell / \bar{\nu}_\ell)$$

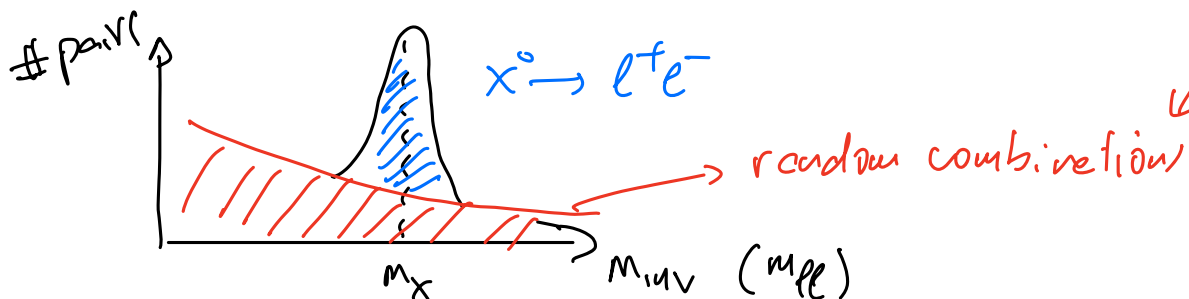
many charged hadrons (a) directly
b) from decay of hadrons)

few leptons from decay of hadrons

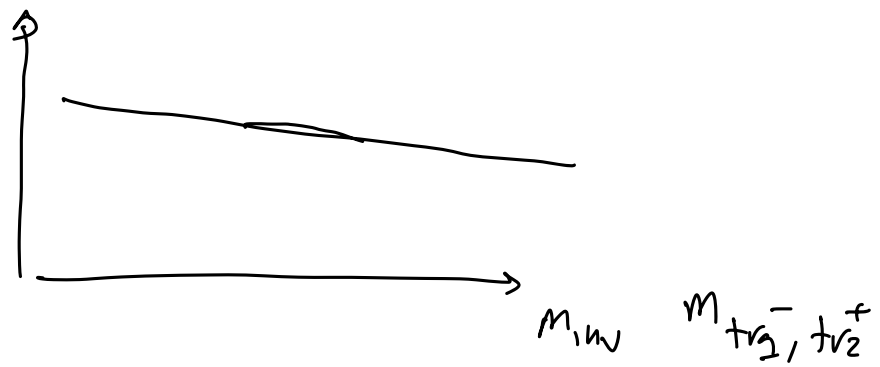
$$\pi^+\pi^-, \pi^+K^-, \ell^+\pi^-, \ell^-\pi^+, \ell^+K^-, \ell^-K^-, e^+e^-$$

$$\pi^+e^+$$

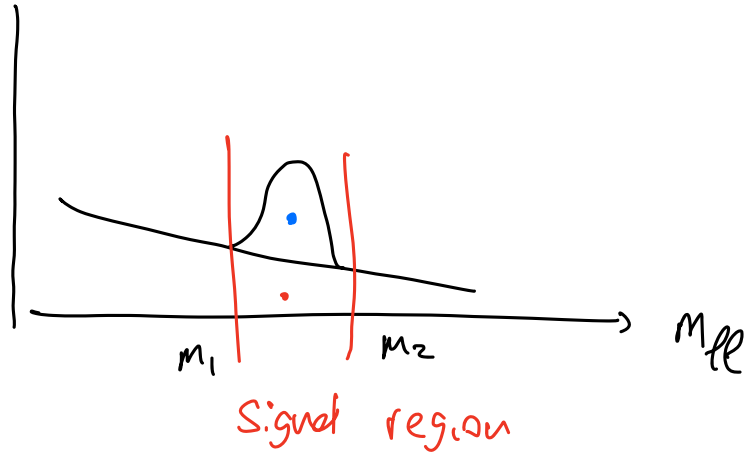
right-sign combinations
if X is neutral.



$$M_{inv} = \sqrt{(\underline{p}_1 + \underline{p}_2)^2} = \sqrt{(E_1 + E_2)^2 - (\underline{p}_1 + \underline{p}_2)^2}$$

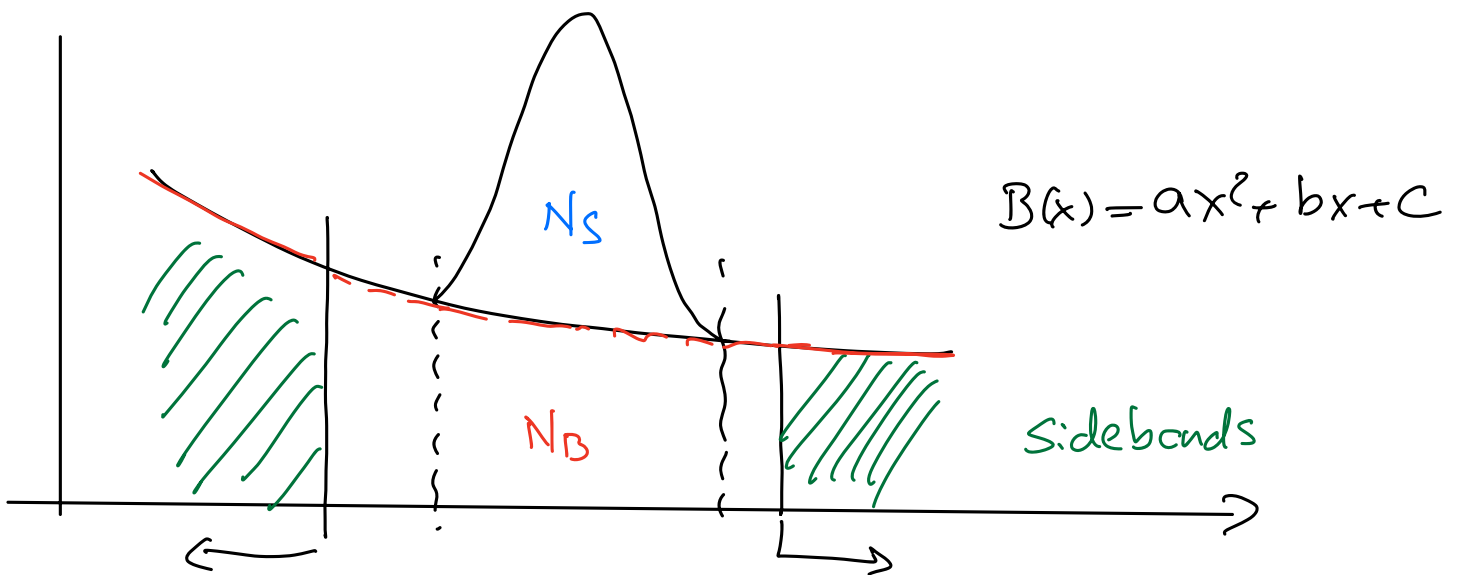


distinguish π, K from e



Noise level:

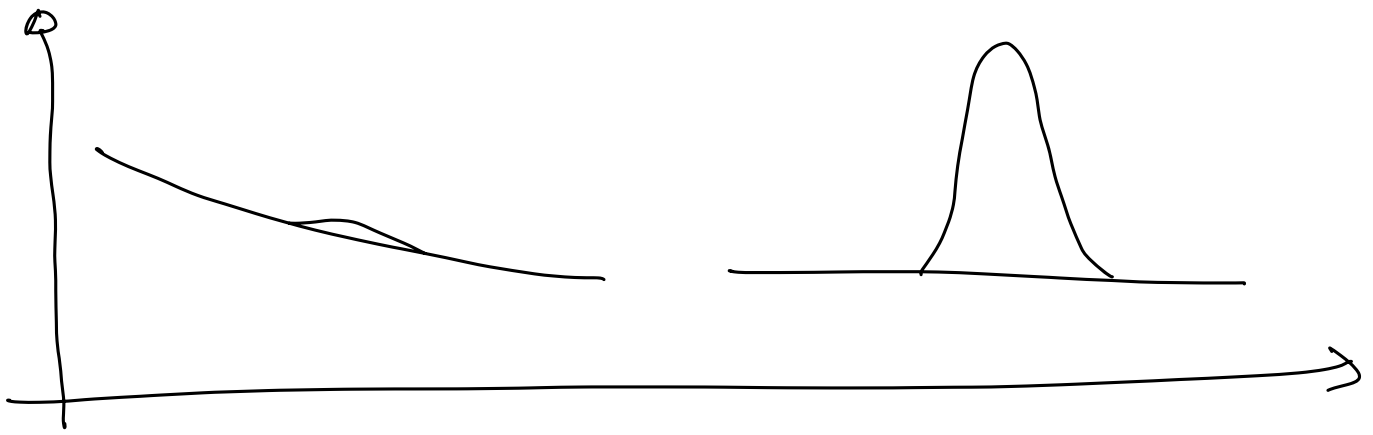
all combinations with $m_1 < m_{\ell\ell} < m_2$



N_B is extrapolated from sideband w/ only background

$$N_B = \int_{m_1}^{m_2} B(x) dx$$

$$N_S = N_{obs} - N_B$$



N_B : follow Poisson distribution. $\delta N_B = \sqrt{N_B}$

significance of signal
$$S = \frac{N_{obs} - N_B \equiv N_S}{\sqrt{N_B}} = \frac{N_S}{\sqrt{N_B}}$$

$S \geq 3$ evidence

$S \geq 5$ observation

